

CertiHeal-Edge: Proofs

This document states and proves Theorem 1, Proposition 1, Theorem 2, Lemma 1, and Lemma 2. All notation is defined in Section 0. Standard results are cited without proof.

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0. Setting and notation

0.1 Service graph, failures, spares, demand

Let $G = (V, E)$ be a finite undirected graph. Let $F \subseteq V$ be the failed-node set and $S \subseteq V \setminus F$ the spare set. Define

$$G_F = G[V \setminus F].$$

For $v \in V$, let $N(v)$ denote its neighbor set in G .

For $v \in F$, let $d_v \in \mathbb{Z}_{\geq 0}$ be the orphaned demand at v , and set

$$d(F) = \sum_{v \in F} d_v.$$

Let $R(F)$ be the set of orphaned roles. Each $r \in R(F)$ has origin $v(r) \in F$ and demand $q_r \in \mathbb{Z}_{\geq 1}$, with

$$d_v = \sum_{r: v(r)=v} q_r.$$

Theorem 1 uses unit roles, $q_r = 1$. Proposition 1 permits arbitrary q_r .

Each live node $v \in V \setminus F$ has capacity $b_v \in \mathbb{Z}_{\geq 0}$. For $s \in S$, b_s is the residual spare capacity. Each live edge $\{v, w\} \in E(G_F)$ has capacity

$$b_{vw} = b_{wv} \in \mathbb{Z}_{\geq 0} \cup \{+\infty\}.$$

Define

$$B_v = N(v) \setminus F, \quad B_F = \left(\bigcup_{v \in F} B_v \right) \setminus S.$$

Then

$$B_F \cap S = \emptyset, \quad \partial = B_F \cup S, \quad I = (V \setminus F) \setminus \partial.$$

0.2 The capacity graph G_F^{cap} (node-split construction)

G_F^{cap} is the directed capacitated graph with the following vertices and arcs.

- Vertices: a super-source σ_F ; one per-node source σ_v for each $v \in F$; for each live node $v \in V \setminus F$ an in-out pair $(v_{\text{in}}, v_{\text{out}})$; and a super-sink τ .
- Source arcs: (σ_F, σ_v) with capacity d_v for each $v \in F$; and $(\sigma_v, w_{\text{in}})$ with capacity $+\infty$ for each $v \in F$ and each $w \in B_v = N(v) \setminus F$. There are no other arcs leaving σ_F or any σ_v , and no arcs enter them.
- Node arcs: $(v_{\text{in}}, v_{\text{out}})$ with capacity b_v for each live v .
- Link arcs: $(v_{\text{out}}, w_{\text{in}})$ and $(w_{\text{out}}, v_{\text{in}})$, each with capacity b_{vw} , for each live link $\{v, w\} \in E(G_F)$.
- Sink arcs: (s_{out}, τ) with capacity $+\infty$ for each spare $s \in S$.

All finite capacities are integers. Write $c(a)$ for the capacity of arc a .

A flow is a function x on arcs with $0 \leq x(a) \leq c(a)$ and flow conservation at every vertex other than σ_F and τ . Its value $|x|$ is the net flow out of σ_F . A σ_F - τ cut is a vertex set X with $\sigma_F \in X$, $\tau \notin X$; its capacity is $c(X) = \sum_{a=(p,q): p \in X, q \notin X} c(a)$. Because the sink arcs are uncapacitated, every finite-capacity cut X contains no s_{out} , $s \in S$; such a cut separates σ_F from every spare and is called a capacitated σ_F - S cut. The certificate is

$$C(F, S) = \max\{|x| : x \text{ a flow in } G_F^{\text{cap}}\}.$$

A recovery configuration is a set $R' \subseteq R(F)$ of unit roles together with, for each $r \in R'$, a simple path π_r in G_F from a node of $B_{v(r)}$ to a spare, such that for every live node v the number of paths using v (as an interior vertex or as an endpoint) is at most b_v , and for every live link $\{v, w\}$ the number of paths traversing it is at most b_{vw} . Its size is $|R'|$, the number of role units served at once.

0.3 The field graph G_F^{field} and the Jacobi iteration

G_F^{field} is the undirected graph G_F with a symmetric conductance $c_{vw} = c_{wv} \in [c_{\min}, c_{\max}]$, $0 < c_{\min} \leq c_{\max} < \infty$, on every live link. The weighted degree of a live node is $d_v^c = \sum_{w \in N_F(v)} c_{vw}$ where $N_F(v)$ is the neighbor set in G_F . A_c is the weighted adjacency matrix, $D_c = \text{diag}(d_v^c)$, and $L_c = D_c - A_c$ is the weighted Laplacian.

The Dirichlet data are

$$g_v = \begin{cases} 1, & v \in B_F, \\ 0, & v \in S. \end{cases}$$

With the ordering (I, ∂) ,

$$u_{\partial} = g, \quad (L_c u)_I = 0,$$

or equivalently

$$L_{II} u_I = -L_{I\partial} g. \tag{0.1}$$

For the CERTIFIED branch, $c = c^0$ and $\lambda = 0$. For $v \in I$,

$$\begin{aligned} u_v^{k+1} &= \frac{1}{d_v^c} \sum_{w \in N_F(v)} c_{vw} u_w^k, \\ u_w^k &= g_w, \quad w \in \partial. \end{aligned} \tag{0.2}$$

Equivalently,

$$u_I^{k+1} = Ju_I^k + h,$$

where

$$J = (D_c^{-1}A_c)_{II}, \quad h = (D_c^{-1}A_c)_{I\partial}g.$$

Let $\rho = \rho(J)$.

For the HEURISTIC branch, let $M > 0$, $\lambda \geq 0$, and

$$p_v = \mathbf{1}_{\{v \in \partial\}}.$$

Then

$$u_v^{k+1} = \frac{\sum_{w \in N_F(v)} c_{vw}u_w^k + Mp_v g_v + \lambda(1 - p_v)\eta_v}{d_v^c + Mp_v}. \quad (0.3)$$

Moreover,

$$J_M = (D_c + MP)^{-1}A_c, \quad P = \text{diag}(p_v).$$

Standing assumption. Every connected component of G_F^{field} containing a vertex of I intersects ∂ . Components disjoint from ∂ are omitted.

0.4 Norms

For $w > 0$, define

$$\|x\|_w = \max_i \frac{|x_i|}{w_i}, \quad \|J\|_w = \max_i \frac{1}{w_i} \sum_j J_{ij}w_j.$$

For a positive diagonal matrix D , define

$$\|x\|_D = (x^\top D x)^{1/2}.$$

1. Theorem 1 (certificate)

Theorem 1 (certificate: exact value, Menger reduction, monotonicity, weak duality). In G_F^{cap} with integer capacities:

- (a) The maximum size of a recovery configuration equals the maximum flow value $C(F, S)$, and a maximum flow can be taken integral.
- (b) $C(F, S)$ equals the minimum capacity of a σ_F - τ cut, which is the minimum capacity of a capacitated σ_F - S cut.
- (c) (Menger reduction.) If $d_v = 1$ for all $v \in F$, $b_v = 1$ for all live v , and links are uncapacitated, then $C(F, S)$ is the maximum number of pairwise vertex-disjoint paths in G_F that start at distinct sets B_v (one path per failed node, starting in its own B_v) and end in S , and equals the minimum number of live vertices whose removal, together with the loss of their source arcs, separates every σ_v from S in G_F^{cap} .
- (d) (Monotonicity.) If $S \subseteq S'$ and the capacities of arcs common to $G_F^{\text{cap}}(S)$ and $G_F^{\text{cap}}(S')$ are not decreased, then $C(F, S) \leq C(F, S')$.
- (e) (Weak duality.) Every feasible flow x in G_F^{cap} satisfies $|x| \leq C(F, S)$; in particular any flow committed by the local rounds is a lower bound on $C(F, S)$.

Proof.

Step 1. Recovery configuration to integral flow. For $r \in R'$, write

$$\pi_r = (w_0, w_1, \dots, w_m), \quad w_0 \in B_{v(r)}, \quad w_m \in S,$$

and define its lift

$$\widehat{\pi}_r = (\sigma_F, \sigma_{v(r)}, w_{0,\text{in}}, w_{0,\text{out}}, \dots, w_{m,\text{in}}, w_{m,\text{out}}, \tau).$$

Set

$$x = \sum_{r \in R'} \mathbf{1}_{\widehat{\pi}_r}.$$

Then x is integral and satisfies flow conservation. Moreover,

$$\begin{aligned} x(\sigma_F, \sigma_v) &= |\{r \in R' : v(r) = v\}| \leq d_v, \\ x(v_{\text{in}}, v_{\text{out}}) &\leq b_v, \\ x(v_{\text{out}}, w_{\text{in}}) &\leq b_{vw}. \end{aligned}$$

Thus x is feasible and $|x| = |R'|$. Hence

$$C(F, S) \geq \max\{|R'| : R' \text{ is a recovery configuration}\}.$$

Step 2. Integral flow to recovery configuration. Let x be an integral feasible flow. If both link arcs associated with a live edge $\{v, w\}$ carry positive flow, cancel the maximal common integer amount on the directed cycle

$$v_{\text{out}} \rightarrow w_{\text{in}} \rightarrow w_{\text{out}} \rightarrow v_{\text{in}} \rightarrow v_{\text{out}}.$$

Cycle cancellation preserves feasibility and $|x|$. Repeating this operation yields a flow with at most one positive direction on each live edge.

Apply integral flow decomposition and discard directed cycles. The resulting flow is a sum of $|x|$ unit $\sigma_F - \tau$ paths. Each such path has the form

$$\sigma_F \rightarrow \sigma_v \rightarrow w_{0,\text{in}} \rightarrow w_{0,\text{out}} \rightarrow \dots \rightarrow w_{m,\text{out}} \rightarrow \tau,$$

with $w_0 \in B_v$ and $w_m \in S$. Its projection to G_F is a simple path from B_v to S .

The source, node, and link capacities give, respectively,

$$\#\{\text{paths assigned to } v\} \leq d_v, \quad \#\{\text{paths using } v\} \leq b_v, \quad \#\{\text{paths using } \{v, w\}\} \leq b_{vw}.$$

Hence these projected paths form a recovery configuration of size $|x|$. Therefore

$$C(F, S) \leq \max\{|R'| : R' \text{ is a recovery configuration}\}$$

for every integral maximum flow.

Step 3. Part (a). Since every flow has value at most

$$d(F) = \sum_{v \in F} d_v,$$

each infinite capacity may be replaced by $d(F)$ without changing the maximum value. The resulting network has integer capacities. By the integrality theorem for maximum flows, it admits an integral maximum flow. Steps 1–2 prove (a).

Step 4. Part (b). The max-flow min-cut theorem gives

$$C(F, S) = \min_X c(X),$$

where X ranges over σ_F - τ cuts. A minimum cut is finite. Hence it contains no s_{out} , $s \in S$, because (s_{out}, τ) has infinite capacity. Thus the finite σ_F - τ cuts are exactly the capacitated σ_F - S cuts relevant to the minimum.

For every feasible flow x and every such cut X ,

$$\begin{aligned} |x| &= \sum_{a \in \delta^+(X)} x(a) - \sum_{a \in \delta^-(X)} x(a) \\ &\leq \sum_{a \in \delta^+(X)} c(a) \\ &= c(X). \end{aligned} \tag{1.1}$$

For a maximum flow, let X be the vertices reachable from σ_F in the residual graph. Then every arc in $\delta^+(X)$ is saturated and every arc in $\delta^-(X)$ carries zero flow. Equality holds in (1.1), proving (b).

Step 5. Part (c). Under $d_v = b_v = 1$ and infinite link capacities, the decomposition in Step 2 gives pairwise vertex-disjoint paths, with at most one path associated with each failed node. Conversely, every such path family defines a feasible unit flow by Step 1.

For a finite cut X , only source arcs and node arcs can leave X . Define

$$F_0 = \{v \in F : \sigma_v \notin X\}, \quad U = \{v \in V \setminus F : v_{\text{in}} \in X, v_{\text{out}} \notin X\}.$$

Then

$$c(X) = |F_0| + |U|.$$

Closure of X under infinite-capacity arcs implies that $G_F - U$ contains no path from B_v to S for $v \in F \setminus F_0$. Hence (F_0, U) is a separator.

Conversely, for such a separator (F_0, U) , let W be the set of vertices reachable in $G_F - U$ from

$$\bigcup_{v \in F \setminus F_0} B_v.$$

Then $W \cap S = \emptyset$. Taking X to contain σ_F , the vertices σ_v for $v \notin F_0$, the split copies of W , and the necessary u_{in} for $u \in U$, gives

$$c(X) \leq |F_0| + |U|.$$

Thus the minimum cut equals the minimum separator size. This is the vertex form of Menger's theorem under node splitting.

Step 6. Part (d). If $S \subseteq S'$, every arc of $G_F^{\text{cap}}(S)$ is present in $G_F^{\text{cap}}(S')$, and no common capacity decreases. Hence every feasible flow for S is feasible for S' with the same value. Therefore

$$C(F, S) \leq C(F, S').$$

Step 7. Part (e). From (1.1), for every feasible flow x and every minimum cut X^* ,

$$|x| \leq c(X^*) = C(F, S).$$

In particular, Proposition 1 gives

$$f_{\text{field}} \leq C(F, S).$$

■

Remark 1.1 (indivisible multi-unit roles). If $q_r > 1$ must be routed unsplittably on one path, every feasible unsplittable routing induces a feasible flow. Hence its served demand is at most $C(F, S)$.

2. Proposition 1 (reservation safety and the witness bound)

2.1 Protocol model

A resource unit is any capacitated arc of G_F^{cap} : the source arc (σ_F, σ_v) (capacity d_v), a node arc $(v_{\text{in}}, v_{\text{out}})$ (capacity b_v ; for a spare this is its slot capacity), or a link arc (capacity b_{vw}). Write $\mathcal{U}$ for the set of units and $\text{cap}(u)$ for the capacity of unit u . Uncapacitated arcs are not units.

An episode has an identifier e ; identifiers are strictly increasing across epochs and never reused. Within episode e , a coordinator computes for each role $r \in R(F)$ a candidate route: a simple path π_r in G_F from a node of $B_{v(r)}$ to a spare (Phase B), or the direct route $\sigma_{v(r)} \rightarrow s$ for an adjacent spare s (Phase 0). The unit set of the route, $\mathcal{U}(r) \subseteq \mathcal{U}$, is the set of capacitated arcs of the lifted path $\hat{\pi}_r$ of Section 1, Step 1: the source arc of $v(r)$, the node arcs of every vertex of π_r , and the link arcs of every edge of π_r in the direction of traversal.

Each unit u keeps, for each key (e, r, u) it has ever seen, a record with state in $\{\text{PREPARED}, \text{COMMITTED}, \text{ABORTED}\}$ and the demand q_r ; a key with no record is FREE. Define the held load

$$\text{held}_u = \sum \{q_r : (e, r, u) \text{ in state PREPARED or COMMITTED}\}.$$

The coordinator keeps, for each role, a state in $\{\text{ROUTED}, \text{PREPARING}, \text{COMMITTING}, \text{ACTIVE}, \text{ABORTED}\}$.

Events are message deliveries or local timeouts. Each event is atomic. The transition rules are:

- (U1) On $\text{RESERVE}(e, r, u)$: if a record for (e, r, u) exists, do nothing except re-send the acknowledgement matching its state (idempotence). Otherwise, if $\text{held}_u + q_r \leq \text{cap}(u)$, create the record in state PREPARED and send $\text{ACK-PREPARE}(e, r, u)$; else send $\text{NACK}(e, r, u)$ and create no record.
- (U2) On $\text{COMMIT}(e, r, u)$: if the record is PREPARED, move it to COMMITTED and send $\text{ACK-COMMIT}(e, r, u)$; if COMMITTED, re-send the acknowledgement; if ABORTED or absent, ignore.
- (U3) On $\text{ABORT}(e, r, u)$, or on expiry of the lease of a PREPARED record: if the record is PREPARED or COMMITTED, move it to ABORTED; if absent, create it in state ABORTED; if ABORTED, do nothing. Records in state ABORTED contribute nothing to held_u and reject every later RESERVE or COMMIT under the same key.
- (C1) Coordinator, role r in ROUTED: send $\text{RESERVE}(e, r, u)$ to all $u \in \mathcal{U}(r)$; go to PREPARING.
- (C2) Coordinator, role r in PREPARING: on receipt of ACK-PREPARE from every $u \in \mathcal{U}(r)$, send $\text{COMMIT}(e, r, u)$ to all $u \in \mathcal{U}(r)$ and go to COMMITTING. On any NACK, or on the coordinator's own lease timeout, send $\text{ABORT}(e, r, u)$ to all $u \in \mathcal{U}(r)$ and go to ABORTED.
- (C3) Coordinator, role r in COMMITTING: on receipt of ACK-COMMIT from every $u \in \mathcal{U}(r)$, activate r and go to ACTIVE (this is the COMMIT-ALL point). On lease timeout before that, send ABORT to all $u \in \mathcal{U}(r)$ and go to ABORTED.
- (C4) A role in ACTIVE or ABORTED never changes state within episode e , and the coordinator sends no further message for it except re-sends of COMMIT to units of an ACTIVE role. A role counts toward f_{field} if and only if it is ACTIVE.
- (C5) Epoch change: when a new coordinator opens episode $e + 1$, it reads the records of every unit; every role of episode e that is ACTIVE stays ACTIVE with its COMMITTED records; every other role of episode e is treated as ABORTED, and $\text{ABORT}(e, r, u)$ is sent to all units of its route. Routing in episode $e + 1$ uses the residual capacities $\text{cap}(u) - \text{held}_u$.

A record is retained longer than the maximum message lifetime. Episode identifiers are strictly increasing and are never reused. Hence late messages remain well defined. The statements below hold for every finite execution prefix consistent with (U1)–(U3) and (C1)–(C5), allowing loss, duplication, and reordering.

2.2 Statement and proof

Proposition 1 (reservation safety). In every reachable global state of the protocol of Section 2.1:

- (i) (Capacity safety.) For every unit u , $\text{held}_u \leq \text{cap}(u)$.
- (ii) (No double-booking.) For every unit u and every (e, r) there is at most one record, and the loads of distinct keys are disjoint parts of held_u ; a record enters PREPARED only from FREE, enters COMMITTED only from PREPARED and only on a COMMIT message carrying its own key, and once ABORTED never leaves ABORTED.
- (iii) (Path atomicity.) If role r is ACTIVE, then every $u \in \mathcal{U}(r)$ holds a COMMITTED record for (e, r, u) ; and a role is ACTIVE only after the coordinator has received ACK-COMMIT from every unit of its route. If the coordinator has issued ABORT for $(e, r, \cdot)$, or a lease of one of r 's PREPARED records has expired, then r is never ACTIVE in episode e and no later message can activate it.

Consequently, at any time, let A be the set of ACTIVE roles across all episodes and let $f_{\text{field}} = \sum_{r \in A} q_r$. Then $x = \sum_{r \in A} q_r \mathbf{1}_{\hat{\pi}_r}$ is a feasible flow in G_F^{cap} for the current failed set F , and hence $f_{\text{field}} \leq C(F, S)$.

Proof. Induct on the event sequence. Initially, $\text{held}_u = 0$ and no record exists.

(U1). An existing key is unchanged. A new PREPARED record is created only if

$$\text{held}_u + q_r \leq \text{cap}(u).$$

Thus capacity safety is preserved and each key is created at most once.

(U2). PREPARED changes to COMMITTED without changing held_u . If ABORT has already been issued, the coordinator is ABORTED by (C4); a delayed COMMIT therefore cannot activate the role. At the unit, either COMMIT is ignored after ABORT or the later ABORT changes the transient COMMITTED record to ABORTED.

(U3). ABORT and PREPARED-lease expiry do not increase held_u , and ABORTED is absorbing. A lease expiry cannot change an ACTIVE role, because every record of an ACTIVE role is COMMITTED. An ABORT is issued only for a non-ACTIVE role; by (C4) that role cannot later become ACTIVE.

(C1), (C2). These rules do not change unit records. A transition to COMMITTING occurs only after ACK-PREPARE from every unit, and a transition to ABORTED is permanent within the episode.

(C3). A role becomes ACTIVE only after ACK-COMMIT from every $u \in \mathcal{U}(r)$. By (U2), each acknowledgement corresponds to a COMMITTED record. No ABORT for that role has been issued before activation. Hence all route records are COMMITTED at activation.

(C5). ACTIVE roles and their COMMITTED records are preserved. All other roles of the previous episode are aborted. Since episode identifiers are strictly increasing, keys from different episodes are distinct.

Thus (i)–(iii) hold in every reachable state.

Let A be the set of ACTIVE roles. Under the epoch model, an ACTIVE role whose route is invalidated by a later node failure is re-orphaned and is no longer ACTIVE. Hence every $r \in A$

has a route in the current G_F . Let $\widehat{\pi}_r$ be its lifted route and define

$$x = \sum_{r \in A} q_r \mathbf{1}_{\widehat{\pi}_r}.$$

By (iii), every ACTIVE role using a capacitated arc u has a distinct COMMITTED record at u . Hence, by (i)–(ii),

$$\begin{aligned} x(u) &= \sum_{\substack{r \in A \\ u \in \mathcal{U}(r)}} q_r \\ &\leq \text{held}_u \\ &\leq \text{cap}(u). \end{aligned}$$

Each $\widehat{\pi}_r$ is a current $\sigma_F - \tau$ path; therefore x is feasible. Its value is

$$|x| = \sum_{r \in A} q_r = f_{\text{field}}.$$

Theorem 1(e) gives

$$f_{\text{field}} \leq C(F, S).$$

■

Remark 2.1. The CERTIFIED, HEURISTIC, and exact-solver routes are committed through the same protocol. Proposition 1 therefore applies to the selected assignment. PREPARING and COMMITTING roles do not contribute to f_{field} .

3. Theorem 2 (fallback branch)

Theorem 2 (fallback branch: per-scenario guarantee). Let c^0 be a fixed symmetric nominal conductance on G_F^{field} with $c_{vw}^0 \in [c_{\min}, c_{\max}]$, $c_{\min} > 0$, and impose the Dirichlet boundary $u = 1$ on B_F , $u = 0$ on S . Then:

- (a) The interior potentials are determined by the partitioned Laplace system $L_{II}u_I = -L_{I\partial}g$, which has a unique solution on every connected component of G_F^{field} that contains a node of $\partial = B_F \cup S$. The solution satisfies $0 \leq u_v \leq 1$ for all v .
- (b) Let $f_0(F, S)$ be the number of demand units that the current-tracing and capacity-reservation procedure (Phase 0, Phase B with $c = c^0$, and Phase C) serves from the nominal solution for the scenario (F, S) . The routes it produces form a recovery configuration, so $f_0(F, S) \leq C(F, S)$.
- (c) The procedure is greedy and $f_0(F, S) < C(F, S)$ is possible (self-stranding).
- (d) Because Phase C selects $f_{\text{out}} = \max(f_\theta, f_0)$ and the nominal branch does not depend on the learned weights θ , the output satisfies $f_{\text{out}}(F, S) \geq f_0(F, S)$ for every scenario (F, S) and every learned policy; equivalently, the inequality holds pointwise in (F, S) .

Proof.

(a). Fix a connected component K intersecting ∂ , and let $I_K = I \cap K$. For $x \in \mathbb{R}^{I_K}$, extend x by zero on ∂ and denote the extension by $\tilde{x}$. Then

$$\begin{aligned} x^\top L_{II}^K x &= \sum_{\substack{\{v,w\} \in E(K) \\ v \in I_K \text{ or } w \in I_K}} c_{vw}^0 (\tilde{x}_v - \tilde{x}_w)^2 \\ &= \sum_{\{v,w\} \subseteq I_K} c_{vw}^0 (x_v - x_w)^2 + \sum_{v \in I_K} \left(\sum_{w \in \partial \cap N_F(v)} c_{vw}^0 \right) x_v^2. \end{aligned} \tag{3.1}$$

The right-hand side is nonnegative. If it is zero, x is constant on each connected component of $G_F^{\text{field}}[I_K]$ and vanishes at every interior vertex adjacent to ∂ . Each such component contains such a vertex by the standing assumption. Hence $x = 0$. Thus L_{II}^K is positive definite, so (0.1) has a unique solution.

For the maximum principle, suppose the maximum value is $M > 1$ and is attained at an interior vertex v . Harmonicity gives

$$u_v = \frac{1}{d_v^c} \sum_{w \in N_F(v)} c_{vw} u_w.$$

Since $u_w \leq M$ for every neighbor w , the equality forces $u_w = M$ for every $w \in N_F(v)$. Propagation along the connected component reaches ∂ , where $g \in \{0, 1\}$, a contradiction. Applying the same argument to a minimum below 0 gives

$$0 \leq u_v \leq 1.$$

(b). Along a Phase-B trace,

$$u_{w_{i+1}} < u_{w_i}.$$

Hence no vertex repeats, and the trace is simple. A successful trace ends at a spare and reserves all source, node, link, and spare resources on its virtual-capacity copy. Phase 0 satisfies the same capacity conditions. Therefore the accepted nominal routes form a recovery configuration, and Theorem 1 yields

$$f_0(F, S) \leq C(F, S).$$

If the nominal branch is committed, Proposition 1 gives the same bound for the committed flow.

(c). Let

$$F = \{v_1, v_2\}, \quad d_{v_1} = d_{v_2} = 1,$$

with live non-spares a, b , spares s_1, s_2 , unit node/spare capacities, infinite link capacities, and

$$E = \{\{v_1, a\}, \{v_1, b\}, \{v_2, a\}, \{a, s_1\}, \{b, s_2\}\}.$$

Take $c^0 \equiv 1$. Then

$$B_{v_1} = \{a, b\}, \quad B_{v_2} = \{a\}, \quad u_a = u_b = 1, \quad u_{s_1} = u_{s_2} = 0.$$

The routes

$$r_1 : b \rightarrow s_2, \quad r_2 : a \rightarrow s_1$$

are vertex-disjoint, so $C(F, S) = 2$. If the tie for r_1 is resolved in favor of a , the greedy procedure uses $a \rightarrow s_1$ for r_1 and leaves no capacity at a for r_2 . Hence

$$f_0(F, S) = 1 < 2 = C(F, S).$$

(d). The nominal branch is independent of θ , and Phase C uses

$$f_{\text{out}} = \max\{f_\theta, f_0\}.$$

Therefore, for every fixed (F, S) ,

$$f_{\text{out}}(F, S) \geq f_0(F, S).$$

Both branch assignments are recovery configurations on their respective virtual-capacity copies. Hence

$$f_0(F, S) \leq C(F, S), \quad f_\theta(F, S) \leq C(F, S),$$

and consequently

$$f_0(F, S) \leq f_{\text{out}}(F, S) \leq C(F, S).$$

■

4. Lemma 1 (synchronous field convergence)

Lemma 1 (field convergence, independent of the value of the learned weights). Consider G_F^{field} with symmetric conductances $c_{vw} \in [c_{\min}, c_{\max}]$, $c_{\min} > 0$, under the standing assumption of Section 0.3. Let $J = (D_c^{-1}A_c)_{II}$ be the interior Jacobi matrix of (0.2) and $\rho = \rho(J)$.

(a) J is nonnegative and substochastic: every row sum is at most 1; the row of an interior node adjacent to ∂ sums to strictly less than 1 (a deficient row), and the row of an interior node with no neighbor in ∂ sums to exactly 1.

(b) $\rho < 1$. More precisely, J is block diagonal over the connected components Q of $G_F^{\text{field}}[I]$, each block J_Q is irreducible with $\rho(J_Q) < 1$, and $\rho = \max_Q \rho(J_Q)$.

(c) The iteration $u_I^{k+1} = Ju_I^k + h$ converges from every initial vector to the unique solution u_I^* of (0.1). The error $e^k = u_I^k - u_I^*$ satisfies $\|e^k\|_{D_c} \leq \rho^k \|e^0\|_{D_c}$, and the asymptotic rate is exactly ρ : $\lim_{k \rightarrow \infty} \|J^k\|^{1/k} = \rho$ in every matrix norm, and $\|J^k e^0\| = \rho^k \|e^0\|$ for a suitable $e^0 \neq 0$. If the initial potentials lie in $[0, 1]$, then $\|e^k\|_\infty \leq \varepsilon$ as soon as

$$k \geq \frac{\log(1/\varepsilon) + \frac{1}{2} \log(|I| \deg_{\max} c_{\max}/c_{\min})}{1 - \rho},$$

which is $O(\log(1/\varepsilon)/(1 - \rho))$ rounds.

(d) (Uniform bound through the conductance ratio and the boundary geometry.) Let ℓ be the maximum over interior nodes of the graph distance in G_F^{field} to ∂ , and suppose a family of shortest paths from each interior node to ∂ can be chosen so that no edge lies on more than κ of them. Then

$$1 - \rho \geq \frac{c_{\min}}{c_{\max}} \cdot \frac{1}{\deg_{\max} \ell \kappa},$$

independently of which conductances in $[c_{\min}, c_{\max}]$ the learned head g_θ produces. On a two-dimensional grid of side D whose Dirichlet set meets every row (for instance when it contains a full side of the grid, or when every row segment of G_F ends at a surviving neighbor of a failed node or at a spare), one can take $\ell \leq D$ and $\kappa \leq D$, so $1 - \rho \geq c_{\min}/(4c_{\max}D^2)$ and the round count is $O(D^2 \log(1/\varepsilon))$; this order is attained, since for the grid with the whole outer frame as Dirichlet set and unit conductance, $1 - \rho = 1 - \cos \frac{\pi}{D+1} = \Theta(1/D^2)$.

(e) The penalized iteration matrix $J_M = (D_c + MP)^{-1}A_c$ of the HEURISTIC branch (0.3), $M > 0$, is likewise nonnegative and substochastic with its boundary rows deficient, and $\rho(J_M) < 1$; the iteration (0.3) converges geometrically to the unique solution of $(L_c + MP)u = MPg + \lambda(I - P)\eta$ with the same conclusions as (c).

Proof.

(a). For adjacent $v, w \in I$,

$$J_{vw} = \frac{c_{vw}}{d_v^c} \geq 0.$$

Moreover,

$$\begin{aligned} \sum_{w \in I} J_{vw} &= \frac{\sum_{w \in I \cap N_F(v)} c_{vw}}{\sum_{w \in N_F(v)} c_{vw}} \\ &= 1 - \frac{\sum_{w \in \partial \cap N_F(v)} c_{vw}}{d_v^c}. \end{aligned}$$

Hence the row sum is at most 1, and it is strictly less than 1 exactly when $N_F(v) \cap \partial \neq \emptyset$.

(b). Order I by the connected components Q of $G_F^{\text{field}}[I]$. Then

$$J = \text{diag}(J_Q)_Q, \quad \rho(J) = \max_Q \rho(J_Q),$$

and every J_Q is irreducible.

Fix Q . By the standing assumption, Q contains a vertex adjacent to ∂ ; hence J_Q has a deficient row. Let

$$J_Q x = \rho(J_Q) x, \quad x > 0,$$

and set

$$m = \max_i x_i, \quad T = \{i : x_i = m\}.$$

For $i \in T$,

$$\rho(J_Q) m = \sum_j J_{ij} x_j \leq m \sum_j J_{ij} \leq m.$$

Thus $\rho(J_Q) \leq 1$. If $\rho(J_Q) = 1$, equality forces every neighbor in Q of every $i \in T$ to lie in T , and the corresponding row sum to equal 1. Irreducibility gives $T = Q$, contradicting the existence of a deficient row. Therefore

$$\rho(J_Q) < 1$$

for every Q , and hence $\rho < 1$.

(c). Since $\rho < 1$,

$$u_I^* = (I - J)^{-1} h$$

is the unique fixed point. For

$$e^k = u_I^k - u_I^*,$$

we have

$$e^{k+1} = J e^k, \quad e^k = J^k e^0.$$

Let D_c denote the restriction of the degree matrix to I , and define

$$\hat{J} = D_c^{1/2} J D_c^{-1/2} = D_c^{-1/2} (A_c)_{II} D_c^{-1/2}.$$

The matrix $\hat{J}$ is symmetric and similar to J . Hence

$$\|\hat{J}^k\|_2 = \rho^k.$$

Therefore

$$\begin{aligned} \|e^k\|_{D_c} &= \|D_c^{1/2} J^k e^0\|_2 \\ &= \|\hat{J}^k D_c^{1/2} e^0\|_2 \\ &\leq \rho^k \|D_c^{1/2} e^0\|_2 \\ &= \rho^k \|e^0\|_{D_c}. \end{aligned}$$

Thus $u_I^k \rightarrow u_I^*$.

Gelfand's formula gives, for every matrix norm,

$$\lim_{k \rightarrow \infty} \|J^k\|^{1/k} = \rho.$$

Since ρ is a Perron eigenvalue of a block attaining the maximum spectral radius, there exists $e^0 \neq 0$ such that

$$J^k e^0 = \rho^k e^0.$$

If $u_I^0, u_I^* \in [0, 1]^I$, then $\|e^0\|_\infty \leq 1$. Also,

$$\begin{aligned} \|e^k\|_\infty &\leq (\min_{v \in I} d_v^c)^{-1/2} \|e^k\|_{D_c} \\ &\leq \rho^k \left(\frac{\sum_{v \in I} d_v^c}{\min_{v \in I} d_v^c} \right)^{1/2} \|e^0\|_\infty \\ &\leq \rho^k \left(\frac{|I| \deg_{\max} c_{\max}}{c_{\min}} \right)^{1/2}. \end{aligned}$$

Thus $\|e^k\|_\infty \leq \varepsilon$ whenever

$$k \geq \frac{\log(1/\varepsilon) + \frac{1}{2} \log(|I| \deg_{\max} c_{\max}/c_{\min})}{1 - \rho},$$

using $\log(1/\rho) \geq 1 - \rho$.

(d). Since $\hat{J}$ is symmetric and $\lambda_{\max}(\hat{J}) = \rho$,

$$\begin{aligned} 1 - \rho &= \lambda_{\min}(I - \hat{J}) \\ &= \min_{x \neq 0} \frac{x^\top L_I x}{x^\top D_c x}. \end{aligned}$$

Extending x by zero on ∂ gives

$$1 - \rho \geq \frac{c_{\min}}{\deg_{\max} c_{\max}} \min_{x \neq 0} \frac{\sum_{\{v,w\}} (\tilde{x}_v - \tilde{x}_w)^2}{\sum_{v \in I} x_v^2},$$

where the edge sum is over edges with at least one endpoint in I .

For each $v \in I$, choose a shortest path

$$\pi_v = (v = y_0, y_1, \dots, y_{\ell_v}), \quad y_{\ell_v} \in \partial, \quad \ell_v \leq \ell.$$

Then

$$\begin{aligned} x_v^2 &= \left(\sum_{i=0}^{\ell_v-1} (\tilde{x}_{y_i} - \tilde{x}_{y_{i+1}}) \right)^2 \\ &\leq \ell_v \sum_{i=0}^{\ell_v-1} (\tilde{x}_{y_i} - \tilde{x}_{y_{i+1}})^2. \end{aligned}$$

If every edge belongs to at most κ selected paths, then

$$\sum_{v \in I} x_v^2 \leq \ell \kappa \sum_{\{v,w\}} (\tilde{x}_v - \tilde{x}_w)^2.$$

Hence

$$1 - \rho \geq \frac{c_{\min}}{c_{\max}} \frac{1}{\deg_{\max} \ell \kappa}.$$

For the stated two-dimensional grid geometry,

$$\ell \leq D, \quad \kappa \leq D, \quad \deg_{\max} = 4,$$

so

$$1 - \rho \geq \frac{c_{\min}}{4c_{\max} D^2}.$$

For the $D \times D$ interior grid with unit conductance and the outer frame Dirichlet,

$$J = \frac{1}{4}(A_{P_D} \otimes I + I \otimes A_{P_D}),$$

and

$$\lambda_{ij}(J) = \frac{1}{2} \left(\cos \frac{i\pi}{D+1} + \cos \frac{j\pi}{D+1} \right).$$

Therefore

$$\rho = \cos \frac{\pi}{D+1}, \quad 1 - \rho = 2 \sin^2 \frac{\pi}{2(D+1)} = \Theta(D^{-2}).$$

(e). For $J_M = (D_c + MP)^{-1}A_c$,

$$\sum_w (J_M)_{vw} = \frac{d_v^c}{d_v^c + Mp_v}.$$

Thus J_M is nonnegative and substochastic, with deficient boundary rows. The block argument of part (b) gives

$$\rho(J_M) < 1.$$

The fixed point satisfies

$$(L_c + MP)u = MPg + \lambda(I - P)\eta.$$

With

$$\hat{J}_M = (D_c + MP)^{-1/2}A_c(D_c + MP)^{-1/2},$$

which is symmetric,

$$\|e^k\|_{D_c+MP} \leq \rho(J_M)^k \|e^0\|_{D_c+MP}.$$

The corresponding iteration count is

$$O\left(\frac{\log(1/\varepsilon)}{1 - \rho(J_M)}\right).$$

Only $c_{vw} \in [c_{\min}, c_{\max}]$ is used. ■

Remark 4.1. The estimate in part (d) depends on the path-congestion parameter κ . The conclusion $1 - \rho = \Theta(D^{-2})$ applies to Dirichlet geometries with $\ell, \kappa = O(D)$; it need not hold for substantially sparser Dirichlet sets.

5. Lemma 2 (partially asynchronous field convergence)

5.1 Asynchronous model

Let $t \in \mathbb{Z}_{\geq 0}$. For each $i \in I$, let $T^i \subseteq \mathbb{Z}_{\geq 0}$ be its update times. If $t \in T^i$,

$$\begin{aligned} u_i(t+1) &= \sum_{j \in I} J_{ij} u_j(\tau_j^i(t)) + h_i, \\ t - \Delta &\leq \tau_j^i(t) \leq t. \end{aligned} \tag{5.1}$$

If $t \notin T^i$, then $u_i(t+1) = u_i(t)$. For $t < 0$, set $u_i(t) = u_i(0)$.

Assume:

- (S1) T^i is infinite for every i ;
- (S2) every interval $\{t, t+1, \dots, t+B-1\}$ contains an element of T^i for every i ;
- (S3) $t - \tau_j^i(t) \leq \Delta$.

Condition (S2) implies (S1). The fixed point is $u_I^* = Ju_I^* + h$.

5.2 Statement and proof

Lemma 2 (asynchronous convergence). Let J be the Dirichlet Jacobi matrix of Lemma 1 with $\rho = \rho(J) < 1$.

(a) In general $\|J\|_\infty = 1$, so J is not a contraction in the unweighted sup-norm; this happens whenever some interior node has no neighbor in ∂ .

(b) There is a positive vector w with $Jw \leq \rho w$ componentwise; consequently $\|J\|_w = \rho < 1$, that is, J is a contraction in the weighted sup-norm $\|\cdot\|_w$ with factor ρ .

(c) Under (S1) and (S3), (5.1) converges to u_I^* from every initial vector. Under (S2) and (S3), if

$$E_0 = \|u_I(0) - u_I^*\|_w,$$

then

$$\|u_I(t) - u_I^*\|_w \leq \rho^m E_0, \quad t \geq m(B + \Delta), \quad m \in \mathbb{Z}_{\geq 0}.$$

Consequently,

$$t \geq (B + \Delta) \left\lceil \frac{\log(E_0/\varepsilon)}{\log(1/\rho)} \right\rceil$$

implies $\|u_I(t) - u_I^*\|_w \leq \varepsilon$. The iteration count is

$$O\left(\frac{(B + \Delta) \log(1/\varepsilon)}{1 - \rho}\right).$$

The same statements hold for J_M with $\rho(J_M)$ in place of ρ .

Proof.

(a). Since $J \geq 0$,

$$\|J\|_\infty = \max_i \sum_j J_{ij}.$$

By Lemma 1(a), a row corresponding to an interior vertex with no boundary neighbor sums to 1. Hence $\|J\|_\infty = 1$ whenever such a vertex exists. For the unit-conductance path

$$s - x_1 - x_2 - x_3 - b, \quad s \in S, \quad b \in B_F,$$

the row of x_2 sums to 1, whereas $\rho = \cos(\pi/4) < 1$.

(b). Write

$$J = \text{diag}(J_Q)_Q$$

over the interior components. For each Q , Perron–Frobenius gives $w_Q > 0$ such that

$$J_Q w_Q = \rho(J_Q) w_Q \leq \rho w_Q.$$

Let w be the concatenation of the w_Q . Then

$$Jw \leq \rho w.$$

Therefore

$$\|J\|_w = \max_i \frac{(Jw)_i}{w_i} \leq \rho.$$

Since every induced norm dominates the spectral radius,

$$\|J\|_w = \rho.$$

Equivalently,

$$\|Jx\|_w \leq \rho \|x\|_w. \tag{5.2}$$

(c). Let

$$e_i(t) = u_i(t) - u_i^*.$$

At an update time $t \in T^i$,

$$e_i(t+1) = \sum_j J_{ij} e_j(\tau_j^i(t)). \quad (5.3)$$

Otherwise $e_i(t+1) = e_i(t)$.

Set

$$T_m = m(B + \Delta),$$

and define

$$P(m) : |e_i(t)| \leq \rho^m E_0 w_i \quad (i \in I, t \geq T_m).$$

For $m = 0$, the bound follows by induction on t . At an update,

$$\begin{aligned} |e_i(t+1)| &\leq \sum_j J_{ij} E_0 w_j \\ &\leq \rho E_0 w_i \\ &\leq E_0 w_i. \end{aligned}$$

Assume $P(m)$. By (S2), for each i there exists

$$t_i \in T^i, \quad T_m + \Delta \leq t_i \leq T_m + \Delta + B - 1.$$

By (S3),

$$\tau_j^i(t_i) \geq t_i - \Delta \geq T_m.$$

Hence

$$\begin{aligned} |e_i(t_i+1)| &\leq \sum_j J_{ij} \rho^m E_0 w_j \\ &\leq \rho^{m+1} E_0 w_i. \end{aligned}$$

The same estimate holds at every later update of i , and e_i is constant between updates. Since

$$t_i + 1 \leq T_m + \Delta + B = T_{m+1},$$

$P(m+1)$ follows. Thus

$$\|e(t)\|_w \leq \rho^m E_0, \quad t \geq m(B + \Delta).$$

For $\varepsilon > 0$, take

$$m = \left\lceil \frac{\log(E_0/\varepsilon)}{\log(1/\rho)} \right\rceil.$$

Using $\log(1/\rho) \geq 1 - \rho$ gives

$$\begin{aligned} m(B + \Delta) &\leq (B + \Delta) \left(1 + \frac{\log(E_0/\varepsilon)}{1 - \rho} \right) \\ &= O\left(\frac{(B + \Delta) \log(1/\varepsilon)}{1 - \rho} \right). \end{aligned}$$

Under (S1) and (S3), define recursively a finite θ_{m+1} by requiring every node to update at least once after $\theta_m + \Delta$. The preceding induction then gives

$$\|e(t)\|_w \leq \rho^m E_0, \quad t \geq \theta_m,$$

so $e(t) \rightarrow 0$.

The proof uses only nonnegativity and a vector $w > 0$ satisfying $Jw \leq \rho w$. Lemma 1(e) gives the same properties for J_M . ■

Remark 5.1. For $B = 1$ and $\Delta = 0$, the estimate reduces to the synchronous bound, up to equivalence of norms.